

Build Job Log: Selective Laser Melting (SLM) of 316L Stainless Steel Flat Washers

Job Information

Job ID: SLM-JOB-2023-001

Date: 2023-10-05

Operator: John Doe

Machine: SLM Solutions 280HL

Part Description: 316L Stainless Steel Flat Washers

Build Platform: Standard 280 x 280 mm

Job Status: Completed Successfully

Overview: This log documents the complete build cycle for 33 units of 316L stainless steel flat washers using selective laser melting technology. The job involved powder preparation via water atomization and subsequent additive manufacturing process.

Material Inputs and Preparation

Raw Materials Consumption

Material Type	Quantity	Batch Number	Notes
X2CrNiMo1712 Stainless Steel for Powder Production	4.11 kg	STL-316L-2309	Virgin material for water atomization
Process Water for Atomization	16.8 kg	H2O-2309-C	Demineralized water, cooling circuit

Powder Preparation Notes: The stainless steel feedstock was processed through the water atomization unit to produce fine powder suitable for SLM processing. Water served as both cooling and fragmentation media during atomization.

Process Parameters and Settings

Water Atomization System Configuration

- Melting energy specification: 2.23 MJ per kg of material processed
- Total material processed through atomizer: 4.11 kg
- Water pressure maintained: 150-200 bar
- Nozzle configuration: Standard 316L settings

SLM Machine Configuration

Parameter	Value	Unit	Notes
Build Chamber Dimensions	280 × 280 × 365	mm	Standard platform
Layer Thickness	30	µm	Standard for 316L
Laser Power (Active)	200	W	Continuous wave
Scan Speed	800	mm/s	Optimized for washers
Hatch Distance	120	µm	Standard spacing
Build Platform Temperature	80	°C	Pre-heat setting

SLM Operational Parameters: - Total processing duration for complete job: 13.38 hours - Machine power consumption during laser operation: 5.5 kW - Machine power consumption during non-laser phases: 3.5 kW - Number of components in single build cycle: 33 pieces

Gas System Operation

Argon Atmosphere Management

Initial Chamber Preparation: - Chamber purging volume: 700 liters - Oxygen level after purging: < 100 ppm - Purge duration: 45 minutes

Continuous Gas Usage During Build: - Argon consumption rate: 54 liters per component - Total components processed: 33 pieces - Gas flow maintained throughout build: 15 L/min - Pressure regulation: 12-15 mbar above atmospheric

Energy Consumption Parameters

Water Atomization Energy Input

- Specific energy requirement for melting: 2.23 MJ per kilogram of material
- Total material mass processed: 4.11 kg
- Atomization system efficiency: Standard operating range

SLM Process Energy Profile

- Build duration: 13.38 hours continuous operation
- Active laser power draw: 5.5 kW
- Standby/system power: 3.5 kW
- Laser duty cycle: Approximately 65% active time

Historical Comparison: Previous similar job (SLM-JOB-2023-000) showed processing time of 14.2 hours with comparable power settings.

Operational Event Log

Timestamp: 2023-10-05 07:30
Event: Machine startup and pre-heat sequence initiated
Status: Normal
Operator: John Doe
Notes: Build platform heating to 80°C, system checks passed

Timestamp: 2023-10-05 08:15
Event: Chamber argon purging commenced
Status: Normal
Oxygen Level: 850 ppm (initial)
Notes: Standard purging procedure initiated

Timestamp: 2023-10-05 09:00
Event: Chamber atmosphere established
Status: Normal
Oxygen Level: 85 ppm
Notes: Build process ready to initiate

Timestamp: 2023-10-05 09:15
Event: Layer deposition and laser melting initiated
Status: Normal
Layer Count: 1 of 2033
Notes: First layer completed successfully, recoater operation normal

Timestamp: 2023-10-05 22:33
Event: Build process completion
Status: Normal
Layer Count: 2033 completed
Notes: All layers processed, system entering cool-down phase

Output Quantification

Manufactured Components

Component Type	Quantity	Total Mass	Quality Status
316L Stainless Steel Flat Washers	33 units	0.61 kg	Passed visual inspection

Material Recovery and Reuse

Material Stream	Quantity	Destination/Use
316L Powder Recovered for SLM Reuse	2.94 kg	Returned to SLM powder handling system
316L Powder for Water Atomization Remelting	0.15 kg	Transferred to powder production area
Recovered Process Water from Atomization	16.4 kg	Returned to water treatment system

Waste Streams

Waste Category	Quantity	Disposition Method
Solid Residue from Water Atomization	0.41 kg	Landfill disposal per standard procedure
Non-recyclable 316L Powder from SLM	0.01 kg	Landfill disposal, contaminated material

System Performance and Maintenance

Equipment Status During Build

SLM Machine Performance: - Laser uptime: 99.8% - Recoater operation: Normal, zero failures - Powder delivery: Consistent flow - Temperature stability: $\pm 2^{\circ}\text{C}$ throughout build

Cooling System: - Chiller operation: Normal - Temperature control: Within specified range - No alarms or interventions required

Maintenance Activities

Pre-Build Maintenance: - Optics cleaning: Completed - Recoater blade inspection: Within tolerance - Powder sieve check: Passed

Post-Build Maintenance: - Build chamber cleaning: Scheduled - Filter replacement: Not required - System calibration: Verified within specifications

Quality Control Notes

In-Process Monitoring: - Layer consistency: Uniform throughout build - Melt pool monitoring: Stable - No delamination observed - Powder bed quality: Consistent

Post-Build Inspection: - All 33 washers visually acceptable - No visible defects or distortions - Dimensions within specified tolerances - Ready for post-processing if required

Operational Summary

The build job completed successfully within the planned timeframe. All system parameters remained within operational specifications throughout the 13.38-hour processing period. The material consumption and output quantities align with expected values for this component geometry and quantity.

Key Performance Indicators: - Build success rate: 100% (33/33 components) - Powder utilization efficiency: Standard range - Gas consumption: As per established parameters - Energy profile: Consistent with machine specifications

Follow-up Actions: - Post-process inspection scheduled - Powder recycling procedure initiated - System ready for next job after standard maintenance

Document generated: 2023-10-06

Approved by: John Doe, Senior Operator

Next scheduled maintenance: 2023-10-20